



Sybil Attacks on Centrality Measures

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Abstract

Centrality measures are fundamental tools for assessing the importance of nodes in a network, with widespread use in security analysis and the study of covert structures. Importantly, these are precisely the domains where participants may have strong incentives to mislead the analysis. In this work, we investigate Sybil attacks on centrality measures, where an adversary creates multiple identities and distributes connections among them to obscure their true importance. We show that computing an optimal hiding strategy is tractable for degree centrality but NP-complete for both closeness and betweenness centralities. Despite this hardness, we draw from the literature on community detection to design heuristic algorithms that perform well in practice. Experiments on real-world covert networks demonstrate that Sybil-based obfuscation can significantly outperform existing hiding strategies. Our results highlight the risks of relying uncritically on centrality-based methods in security-sensitive applications.

CCS Concepts

- Security and privacy → Social network security and privacy;
- Human-centered computing → Social networks.

Keywords

centrality measure; sybil attack; social network; hiding in networks

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The source code of this paper has been made publicly available at <https://doi.org/10.5281/zenodo.18304796>.

1 Introduction

Centrality measures are metrics used in network analysis to quantify the relative importance of a node within a network [5]. They are widely applied in fields such as sociology, biology, and urban planning to identify key players, bottlenecks, and structural vulnerabilities in complex systems. In particular, centrality measures play a critical role in security and cybersecurity domains, where they

are used to identify leaders in covert networks [23], fraudulent accounts in electronic marketplaces [8], and crucial hosts in computer networks [22]. Consequently, nodes in these domains have strong incentives to manipulate centrality measures to obscure their true influence or importance, a concern that is especially relevant for online platforms, the Web, and privacy-sensitive systems.

A prominent attack type in networked domains is the Sybil attack [15], in which an adversary illegitimately creates multiple fake identities (network nodes) in a distributed system. Traditionally, Sybil attacks considered in the literature aim to spread misinformation or gain disproportionate influence over network processes. Historical examples illustrate this concept in the domain of covert networks: Juan Pujol García created a network of fictitious agents to feed Nazi intelligence false information during World War II [18]; Sidney Reilly, a spy for multiple powers and inspiration for James Bond, maintained multiple false identities [24]; and Ferdinand Demara, “The Great Impostor”, adopted at least a dozen different identities throughout his criminal career [11].

Prior research on attacks targeting centrality measures has largely focused on edge rewiring, i.e., adding or removing network connections [10, 13, 34, 35]. Separately, Sybil attack studies have concentrated on detecting manipulations in trust-based systems [41, 42], often overlooking the threat posed to centrality-based analysis. Our work closes this gap by developing a theoretical and empirical understanding of how Sybil attacks can systematically distort centrality measures, with relevance to online social networks, Web platforms, and security-related applications.

To address this challenge, we formalize the problem of hiding a node’s importance through Sybil attacks as an optimization problem over the network structure. Specifically, we investigate how an adversary can distribute connections among multiple forged identities to minimize the centrality of the original node. We analyze the computational complexity of this problem for three widely used centrality measures, namely degree, closeness, and betweenness, to determine the theoretical limits of manipulation. Based on this formalization, we design a set of heuristic algorithms inspired by community detection techniques to approximate optimal Sybil-based hiding strategies and evaluate their effectiveness empirically on both synthetic and real-world networks.

Our contributions are threefold. First, we provide a formal definition of Sybil-based hiding attacks on centrality measures and analyze their computational complexity, showing that the problem is polynomially solvable for degree centrality but NP-complete for both closeness and betweenness centralities. Second, we propose a family of heuristics that enable practical Sybil-based obfuscation of node importance. Third, through experiments on real-world covert networks, we demonstrate that Sybil-based approaches outperform existing edge-rewiring methods in reducing node visibility



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across multiple centrality metrics. To our knowledge, this represents the first systematic study of how Sybil attacks can directly target centrality-based analyses.

The implications of our findings extend beyond algorithmic interest. In security-sensitive domains, such as intelligence analysis, online privacy, and cybersecurity, centrality measures are often used to identify influential nodes or monitor network activity. Our results show that such analyses can be systematically deceived through multi-identity manipulation, calling for careful interpretation of centrality-based evidence. More broadly, this work highlights the need for robustness, identity verification, and cross-layer validation in networked systems, particularly those operating on the Web, where user anonymity and multiple accounts can be exploited to evade detection.

The remainder of the paper is organized as follows. Section 2 reviews related work. Section 3 introduces network notations and centrality measures. Section 4 formally defines the evader's problem and investigates its computational complexity. Section 5 presents our empirical evaluation on synthetic and real-life networks. Finally, Section 6 discusses implications and directions for future research.

2 Related Work

Douceur [15] was the first to introduce the concept of the Sybil attack, named after the pseudonym of a woman with dissociative identity disorder described by Schreiber [30]. In his seminal work, Douceur not only defined the attack but also explored potential countermeasures, suggesting that systems could limit an attacker's ability to create multiple identities by imposing computational costs on users. This idea anticipated the proof-of-work class of cryptographic protocols [20].

Over the years, a variety of mechanisms have been proposed to defend networks against Sybil attacks. SybilGuard [42] and Sybil-Limit [41] offer probabilistic guarantees of low false acceptance rates for Sybil nodes. Both approaches rely on the assumption that Sybil identities typically form a tightly connected cluster with only a few edges linking them to honest nodes. Their verification process involves performing random routes (walks with fixed edge permutations) from both the verifier and the suspected node, accepting the node only if the routes intersect. Similarly, SybilDefender [39, 40] leverages random walks and assumes that a small graph cut separates the honest and Sybil regions of the network. The SybilInfer algorithm [12] instead adopts a Bayesian approach, classifying nodes as Sybil or honest based on probabilistic inference over the network's structure. Viswanath et al. [32] provide a comprehensive review of Sybil defense mechanisms, highlighting that many rely on the principles of community detection, and that networks with strong community structure among honest nodes are particularly vulnerable to strategic Sybil attacks. As we demonstrate in this work, assumptions made by this class of defense algorithms does not hold in the context of centrality manipulation: for an attacker attempting to reduce their apparent importance, maintaining a cohesive Sybil cluster is in fact detrimental. This limits the effectiveness of traditional Sybil defenses in scenarios where the attacker's objective is to distort centrality-based analyses rather than to infiltrate the network.

Beyond Sybil detection, the broader network science literature has explored various methods of hiding from algorithmic analysis. Researchers have proposed evasion strategies targeting link prediction algorithms [36, 43], node similarity measures [14], and source detection techniques [33]. Several studies have specifically examined the manipulation of centrality measures, both to increase perceived influence [10] and to decrease it [35]. Further works extended this inquiry to collective hiding, where the goal is to lower the centrality rankings of a group of nodes [13, 34] in order to obscure the identity of organizational leaders. However, existing methods are almost exclusively based on edge rewiring, i.e., adding or removing links in the network.

In contrast, our work introduces and systematically studies a new class of Sybil-based attacks on centrality measures, where adversaries employ multiple false identities to obfuscate their importance without directly manipulating the existing edges of the network. This perspective bridges two previously disjoint research areas, i.e., centrality manipulation and Sybil-based deception, offering a unified framework for understanding how multi-identity strategies can undermine centrality-driven analyses in both social and security-critical networks.

3 Preliminaries

This section introduces the notation and definitions used throughout the paper, including the basic network model and the centrality measures analyzed in our work. A summary of all symbols is provided in Table 2 in the appendix.

3.1 Basic Network Notation

Let $G = (V, E)$ denote a network, where V is the set of nodes with $|V| = n$, and $E \subseteq V \times V$ is the set of edges with $|E| = m$. Throughout this work, we consider *undirected* networks, meaning that the edges (v, w) and (w, v) are equivalent. We also assume that the network contains no self-loops, i.e., $\forall v \in V, (v, v) \notin E$.

Let $N_G(v)$ denote the *neighborhood* of node v in G , defined as $N_G(v) = \{w \in V : (v, w) \in E\}$. The *degree* of node v is then given by $\delta_G(v) = |N_G(v)|$. We denote by $d_G(v, w)$ the *distance* between nodes v and w , defined as the number of edges in the shortest path connecting them. If no path exists, we set $d_G(v, w) = \infty$. The set of nodes *reachable* from v is denoted by $K_G(v) = \{w \in V : d_G(v, w) < \infty\}$. Finally, let $\pi_G(v, w)$ denote the set of all shortest paths between v and w in G .

When the underlying graph G is clear from context, we omit the subscript, writing $d(v, w)$, $K(v)$, or $\pi(v, w)$ for brevity. We also slightly abuse notation by allowing nodes and edges to be used directly in set operations as singletons. For example, $v \times V$ denotes $\{v\} \times V$, and $E \setminus (v, w)$ denotes $E \setminus \{(v, w)\}$.

3.2 Centrality Measures

A *centrality measure* is a function $c(G, v)$ that quantifies the relative importance of a node $v \in V$ in a network G [5]. In this work, we focus on four classical centrality measures: degree, closeness, betweenness, and eigenvector centrality.

- **Degree centrality** [31] measures the importance of a node based on its number of direct connections:

$$c_{deg}(G, v) = \delta_G(v).$$

- **Closeness centrality** [6] quantifies the importance of a node based on its proximity to all other nodes in the network. We adopt the formulation proposed by Wasserman [37], which accounts for disconnected graphs:

$$c_{clo}(G, v) = \frac{|K_G(v)| - 1}{n - 1} \cdot \frac{|K_G(v)| - 1}{\sum_{w \in K_G(v) \setminus v} d_G(v, w)}.$$

- **Betweenness centrality** [1, 17] measures the fraction of shortest paths between pairs of nodes passing through v :

$$c_{bet}(G, v) = \sum_{u, w \in K_G(v) \setminus v} \frac{|\{p \in \pi(u, w) : v \in p\}|}{|\pi(u, w)|}.$$

- **Eigenvector centrality** [9] defines a node's importance recursively, based on the importance of its neighbors. Let A denote the adjacency matrix of G , and let λ^* be its largest eigenvalue. Then:

$$c_{eig}(G, v) = x_v, \quad \text{where } Ax = \lambda^* x.$$

4 The Problem and Its Complexity

We now formally define the problem faced by an evader performing a Sybil attack on a centrality measure. We begin by introducing the notion of a Sybil attack as used in our analysis.

DEFINITION 1 (SYBIL ATTACK). Let $G = (V, E)$ be a network, and let $v^* \in V$ denote the node performing the Sybil attack. A Sybil attack is defined as a pair $\mathcal{A} = (Q, \mathcal{E})$ such that $Q \cap V = \emptyset$ and $\mathcal{E} \subseteq (Q \times Q) \cup (Q \times N_G(v^*))$.

The network resulting from the attack is denoted by $\mathcal{A} \cdot G = (\mathcal{A} \cdot V, \mathcal{A} \cdot E)$, where:

- $\mathcal{A} \cdot V = V \setminus v^* \cup Q$,
- $\mathcal{A} \cdot E = E \setminus (v^* \times V) \cup \mathcal{E}$.

We denote by $\emptyset$ the empty attack that leaves the network unchanged, i.e., $\emptyset \cdot G = G$.

Intuitively, after performing the Sybil attack, the evader v^* is replaced by a set of nodes Q . These nodes, all controlled by v^* , may connect both to one another (the $Q \times Q$ component of $\mathcal{E}$) and to the former neighbors of v^* (the $Q \times N_G(v^*)$ component of $\mathcal{E}$). Throughout this paper, we refer to the nodes in Q as *Sybil*s, and to the nodes in $N_G(v^*)$ as *alters*, following the terminology of ego-network analysis [2].

We can now define the decision problem that constitutes the focus of our study.

DEFINITION 2 (CENTRALITY SYBIL ATTACK PROBLEM). An instance of the problem is defined by a tuple (G, v^*, c, b, θ) , where:

- $G = (V, E)$ is a network,
- $v^* \in V$ is the evader node,
- c is a centrality measure,
- $b \in \mathbb{N}$ is the hiding budget,
- $\theta \in \mathbb{N}$ is the safety threshold.

The objective is to identify a Sybil attack $\mathcal{A} = (Q, \mathcal{E})$ such that $|Q| = b$ and the following conditions hold:

- (1) $\forall \varrho \in Q \exists v \in N_G(v^*) d_{\mathcal{A} \cdot V}(\varrho, v) < \infty$,

Centrality measure	Centrality Sybil Attack
Degree	P (Theorem 1)
Closeness	NP-complete (Theorem 2)
Betweenness	NP-complete (Theorem 3)

Table 1: Summary of our computational complexity results.

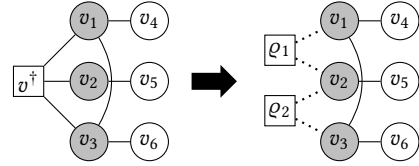


Figure 1: Example of a network in which a single alter needs to be connected with more than one Sybil for $\theta = 3$, and $b = 2$. The alters are grey, the evader and the Sybils are the square nodes, and the edges added to the Sybils are dotted.

$$(2) \forall v \in N_G(v^*) \exists \varrho \in Q (\varrho, v) \in \mathcal{E},$$

$$(3) \exists H \subset \mathcal{A} \cdot V (|H| \geq \theta \wedge \forall h \in H \forall \varrho \in Q c(\mathcal{A} \cdot G, h) > c(\mathcal{A} \cdot G, \varrho)).$$

In other words, the goal is to construct a Sybil attack that preserves direct communication with all alters while ensuring that none of the Sybils appear among the top θ nodes in the centrality ranking. We say that a non-Sybil node v *contributes toward the safety threshold θ* if its centrality exceeds that of all Sybil nodes.

This formulation captures several practical scenarios. In covert networks, a leader may wish to maintain operational ties with subordinates while avoiding detection as the organization's central figure. Similarly, in technological settings, a critical server might distribute its functionality across multiple machines to retain connectivity while reducing its prominence as a potential attack target. The safety threshold θ quantifies the degree of concealment, consistent with prior literature [13, 34]. An alternative formulation could set a fixed centrality value below which all Sybils must fall [35]. However, in most realistic cases, the evader's concern lies in their relative ranking—avoiding being among the top nodes—rather than achieving a specific numerical centrality.

Table 1 summarizes our findings regarding the computational complexity of the Centrality Sybil Attack problem.

Let us begin by considering the case of a Sybil attack on degree centrality, the simplest among the centrality measures studied. At first glance, one might expect the optimal strategy to be straightforward: connect each alter to a distinct Sybil, thereby keeping the degree of every Sybil low. However, this intuition does not always yield the best result. Consider the network illustrated on the left-hand side of Figure 1. Connecting node v_2 to only one of the Sybils fails to satisfy the safety threshold θ , whereas linking it to both Sybils successfully meets the required condition. As we show below, the problem admits an efficient polynomial-time solution, which can be obtained through a greedy algorithm.

THEOREM 1. Algorithm 1 solves the Centrality Sybil Attack problem for the degree centrality in polynomial time.

Algorithm 1 Solving the problem for degree centrality.

Input: The network $G = (V, E)$, the evader $v^\dagger \in V$, the hiding budget $b \in \mathbb{N}$, and the safety threshold $\theta \in \mathbb{N}$.

Output: A solution $(Q, \mathcal{E})$ to the instance $(G, v^\dagger, c_{deg}, b, \theta)$ of the Centrality Sybil Attack problem or $\perp$ if there is no solution.

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1: if  $|\{v \in V : \delta_G(v) > \delta_G(v^\dagger)\}| \geq \theta$  then
2:   return  $\emptyset$ 
3: for  $\kappa \in \langle \lceil \delta_G(v^\dagger)/b \rceil, \dots, \delta_G(v^\dagger) \rangle$  do
4:   if  $\text{TESTSYBILDEGREE}(\kappa) \neq \perp$  then
5:     return  $\text{TESTSYBILDEGREE}(\kappa)$ 
6: return  $\perp$ 
7: function  $\text{TESTSYBILDEGREE}(\kappa)$ 
8:    $x \leftarrow |\{v \in V \setminus (N_G(v^\dagger) \cup v^\dagger) : \delta_G(v) > \kappa\}|$ 
9:    $(Q, \mathcal{E}) \leftarrow (\{\varrho_1, \dots, \varrho_b\}, \emptyset)$ 
10:   $i \leftarrow 1$ 
11:  for  $v \in N_G(v^\dagger)$  sorted decreasingly based on  $\delta_G(v)$  do
12:    if  $x \geq \theta \vee \delta_G(v) > \kappa$  then
13:       $y \leftarrow 1$ 
14:    else
15:       $y \leftarrow \kappa + 1 - \delta_G(v)$ 
16:      if  $y > b$  then
17:        return  $\perp$ 
18:      for  $j = 1, \dots, y$  do
19:         $\mathcal{E} \leftarrow \mathcal{E} \cup (\varrho_i, v)$ 
20:         $i \leftarrow (i \bmod b) + 1$ 
21:       $x \leftarrow x + 1$ 
22:  if  $x \geq \theta \wedge |\mathcal{E}| \leq bk$  then
23:    return  $(Q, \mathcal{E})$ 
24:  return  $\perp$ 

```

PROOF. The key parameter determining the solution is the maximum degree of the Sybil nodes, denoted by κ . Only nodes whose degree exceeds κ contribute to satisfying the safety threshold θ . The main idea behind Algorithm 1 is to enumerate all feasible values of κ and test whether any of them lead to a valid solution. Throughout the analysis, we assume that $b < n$, since otherwise the optimal strategy is trivial—each alter can be connected to a distinct Sybil.

Lines 1–2 verify whether any modification to the network is required, or if the safety threshold θ is already satisfied. The smallest feasible value of κ is $\lceil \delta_G(v^\dagger)/b \rceil$, as each alter must be connected to at least one Sybil. The loop in line 3 iterates over all possible values of κ , invoking the function TESTSYBILDEGREE for each candidate.

We now show that if a valid solution exists for a given κ , it will be identified by TESTSYBILDEGREE . To satisfy θ , the algorithm considers contributions from two types of nodes: those not connected to $v^\dagger$ (counted in line 8) and the alters themselves. The variable i (line 10) is used to distribute edges in $\mathcal{E}$ evenly across the Sybil set Q , ensuring that the maximum degree among Sybils is minimized.

For a fixed κ , at most $b\kappa$ edges can be added to $\mathcal{E}$. The number of edges that an alter v requires to contribute to θ is $\max(1, \kappa + 1 - \delta_G(v))$, which decreases monotonically with $\delta_G(v)$. Consequently, the loop in line 11 processes alters in decreasing order of their degrees. If θ has already been met or v already contributes to it, then v only needs to be connected to a single Sybil (lines 12–13). Otherwise, the algorithm determines the number of additional edges

that must be created (line 15). If adding these edges is impossible given the remaining budget, no valid solution exists for the current κ (lines 16–17). Because alters are processed in order of decreasing degree, none of the subsequent alters will be able to contribute to θ either. The edge additions are performed in lines 18–20, and line 21 records that v contributes to the safety threshold. If a valid configuration is found, it is returned in lines 22–23; otherwise, the algorithm outputs $\perp$. $\square$

The time complexity of a single call to TESTSYBILDEGREE is $O(n(\log n + b))$. Sorting the alters in line 11 requires $O(n \log n)$ time, while each of the $O(n)$ iterations of the loop in line 11 incurs a cost of $O(b)$, driven primarily by the operations in lines 18–20. Consequently, the total time complexity of Algorithm 1 is $O(n^2(\log n + b))$, and its space complexity is $O(n^2)$, dominated by the storage requirements for $\mathcal{E}$.

THEOREM 2. *The Centrality Sybil Attack problem is NP-complete for the closeness centrality.*

THEOREM 3. *The Centrality Sybil Attack problem is NP-complete for the betweenness centrality.*

The proofs of Theorems 2 and 3 can be found in the appendix.

5 Empirical Evaluation

In this section, we present the experimental evaluation of Sybil attacks on centrality measures, focusing on their practical feasibility and effectiveness. We conduct experiments on both synthetic and real-world covert networks to assess how the attacks influence different types of network structures.

5.1 Network Datasets

Our evaluation employs two categories of networks: synthetic models, which allow controlled analysis of structural properties, and real-world covert networks, which illustrate practical implications in realistic settings.

We generate synthetic networks using three well-established random graph models:

- **Erdős-Rényi networks** [16]—classical random graphs in which each possible edge is present with a fixed probability;
- **Watts-Strogatz networks** [38]—networks exhibiting the small-world property, characterized by short average path lengths and high clustering;
- **Barabási-Albert networks** [4]—scale-free networks generated through preferential attachment mechanism, resulting in a power-law degree distribution.

To complement the synthetic analysis, we use several datasets describing real-world covert organizations:

- **November 17**—members of the Revolutionary Organization 17 November, a Marxist guerrilla group active in Greece;
- **Embassy 2004**—individuals involved in the 2004 bombing of the Australian Embassy in Jakarta;
- **Bali 2002**—individuals connected to the 2002 Bali bombing orchestrated by Jemaah Islamiyah;
- **WTC Hamburg**—associates of the Al-Qaeda sleeper cell based in Hamburg prior to the 9/11 attacks;
- **Christmas Eve**—individuals linked to the 2000 Christmas Eve bombings in Jakarta;

- **Madrid 2004**—members associated with the 2004 Madrid train bombings.

All datasets are obtained from the John Jay & ARTIS Transnational Terrorism Database [3]. Table 3 in the appendix summarizes their basic structural properties.

5.2 Experimental Procedure

For each network $G = (V, E)$, we consider as potential evaders only nodes with degree at least 5, since nodes with very few alters offer limited opportunities for strategic concealment.

For every selected evader, we record its ranking position according to the chosen centrality measure in the original network G . We then compare this value with the ranking of the most exposed Sybil—that is, the Sybil node with the highest centrality value—after performing the attack according to a given strategy. In cases where multiple nodes share the same centrality value, we assign them the same ranking position, defined as one plus the number of nodes with strictly higher centrality (e.g., all nodes with the maximum centrality value receive rank one).

Results for synthetic networks are reported as averages over 100 independent network instances, with 10 evaders randomly selected from the set of eligible nodes. Unless stated otherwise, all synthetic networks consist of 1,000 nodes with an average degree of 10. For the Watts-Strogatz model, we use a rewiring probability of $\beta = \frac{1}{4}$. Results for real-world covert networks are averaged over the 10 nodes with the highest degrees, which typically correspond to the most central individuals in each network.

5.3 Simulation Results

We now present the results of our simulations, organized into three experimental scenarios.

5.3.1 Density of Connections. The fundamental challenge in performing a Sybil attack on centrality measures lies in determining how the Sybils should connect both to one another and to the alters. To build an initial understanding of this process, we conduct a series of experiments in which these two types of edges are added uniformly at random (ensuring that each alter is connected to at least one Sybil). Our goal is to analyze how the density of these connections affects the effectiveness of hiding across different centrality measures. Here, we set the number of Sybils to $b = 10$.

Figure 2 summarizes the results. Overall, adding excessive connections among Sybils reduces the effectiveness of the hiding strategy. This finding contrasts with the commonly observed tendency in traditional Sybil attacks, where Sybil nodes typically form tightly interconnected clusters.

The effect of connections between Sybils and alters is less straightforward. When connections among Sybils are sparse, it is generally advantageous to also minimize links to alters (as seen in the leftmost columns of all heatmaps). However, when maintaining some degree of internal connectivity among Sybils is necessary (e.g., in the case of distributed computing systems requiring inter-Sybil communication), the patterns vary by centrality measure. For degree, closeness, and eigenvector centralities, maintaining few Sybil–alter connections remains preferable. In contrast, for betweenness centrality, a higher density of Sybil–alter connections proves beneficial. This behavior likely stems from the fact that additional connections

enable shortest paths between Sybils to pass through alters rather than other Sybils, thereby lowering the Sybils’ betweenness scores.

Notably, the overall trends remain consistent across the different network generation models, suggesting that the type of centrality measure considered has a greater influence on the attack dynamics than the specific structural characteristics of the underlying network.

5.3.2 Heuristic Approaches. Building on the insights from the density analysis, we now evaluate several heuristic Sybil attack strategies and examine how their effectiveness depends on the evader’s initial position in the centrality ranking. In these simulations, we use $b = 5$ Sybils and consider the following strategies:

- **Random**—as a baseline, each alter is connected to a Sybil chosen uniformly at random.
- **Optimal degree**—the algorithm for optimal hiding against degree centrality (Algorithm 1).
- **Community-based strategies**—a family of heuristics guided by community detection. For each strategy, we apply a community detection algorithm to the subgraph induced by the alters in the original network. If possible, we partition the alters into a number of communities equal to the number of Sybils, connecting each community to a distinct Sybil. If the algorithm does not permit specifying the number of communities, we assign them to Sybils in a round-robin manner. We consider the following algorithms: **Girvan–Newman** [26], **Leading Eigenvector** [25], **Walktrap** [27], **Infomap** [29], **Louvain** [7], and **Spinglass** [28].

To simulate a realistic scenario where some inter-Sybil connections are unavoidable, we also connect the Sybils in a ring topology. This introduces minimal connectivity among Sybils without creating excessive internal density. For the optimal degree algorithm, we adjust for this setup by accounting for the additional degree incurred by the two ring edges when calculating κ .

Figure 3 presents the simulation results. As shown, the magnitude of ranking change strongly depends on the evader’s initial position. Evaders occupying top-ranking positions—those with the greatest incentive to conceal their importance—achieve the largest drops in ranking following a Sybil attack. In contrast, nodes that are initially less central experience smaller absolute ranking changes, though they still tend to improve their relative anonymity compared to high-ranking nodes.

Regarding the effectiveness of specific heuristics, the random strategy performs poorly in most settings, often increasing the evader’s visibility. Community-based approaches yield comparable results, except for infomap, which sometimes exposes the evader even further. Notably, the optimal degree algorithm consistently performs well across all four considered centrality measures, suggesting it is a robust choice when the attacker does not know which centrality metric is being used.

Among the centrality measures, betweenness once again displays distinct behavior. In many cases, Sybil attacks actually make the evader more visible (as indicated by points below the diagonal line). This observation implies that prioritizing betweenness centrality may offer a natural defense against Sybil-based obfuscation.

Finally, the structure of the underlying network also plays a key role. The hiding process is least effective in scale-free networks

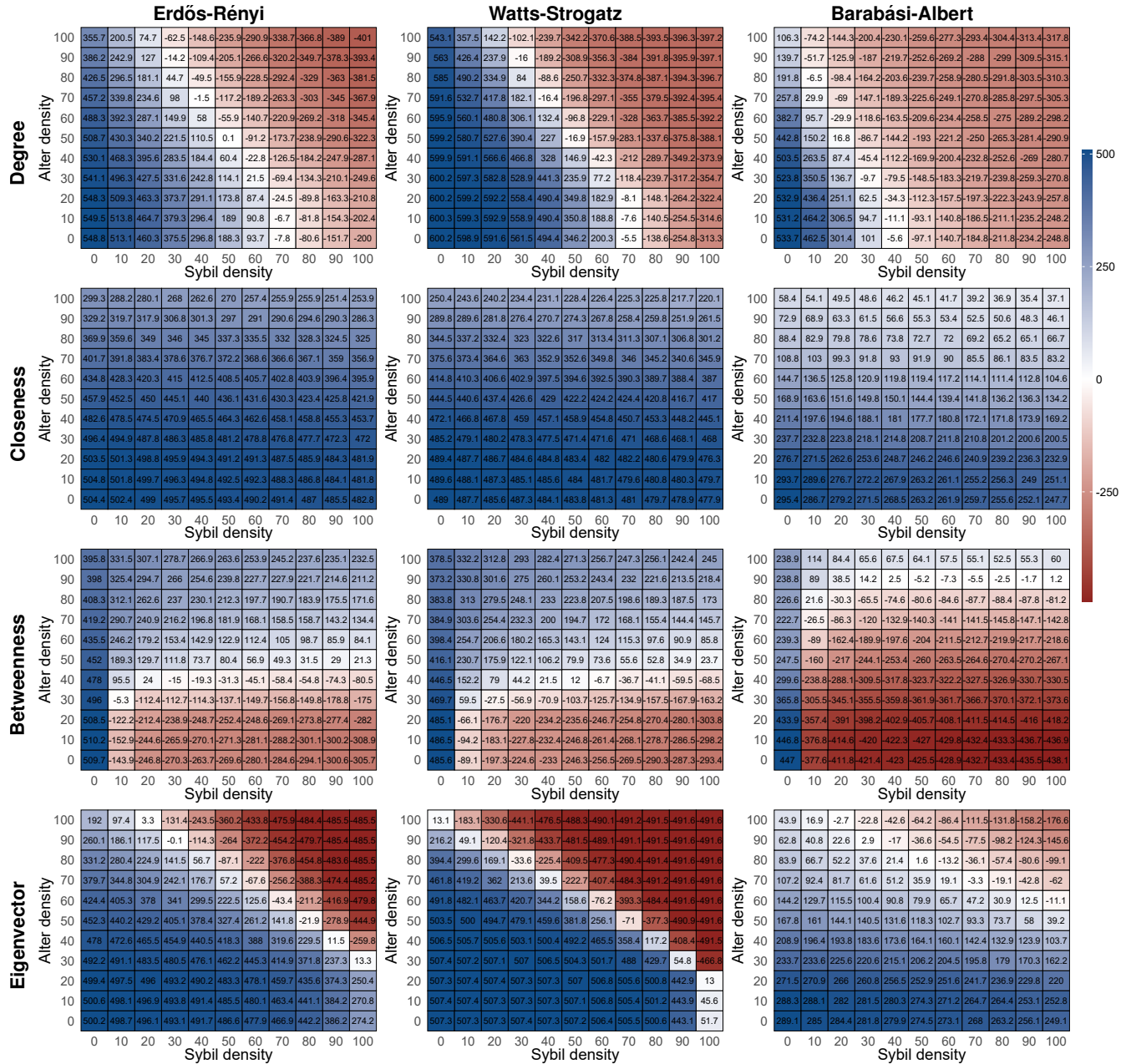


Figure 2: Effects of varying the density of connections between Sybils and between Sybils and alters. Each row corresponds to a different centrality measure, while each column represents results for a different network generation model. In each heatmap, the x -axis shows the density of connections among Sybils, and the y -axis shows the density of connections between Sybils and alters, both expressed as a percentage of the maximum possible density. Each cell indicates the difference between the centrality ranking of the most exposed Sybil and that of the original evader, higher values indicate more effective hiding.

generated using the Barabási-Albert model, where results cluster near the diagonal line. The presence of dominant hubs—likely occupying top centrality ranks—makes hiding particularly difficult.

Conversely, the Erdős-Rényi networks show the greatest susceptibility to Sybil attacks, though they are also the least realistic of the three synthetic network models.

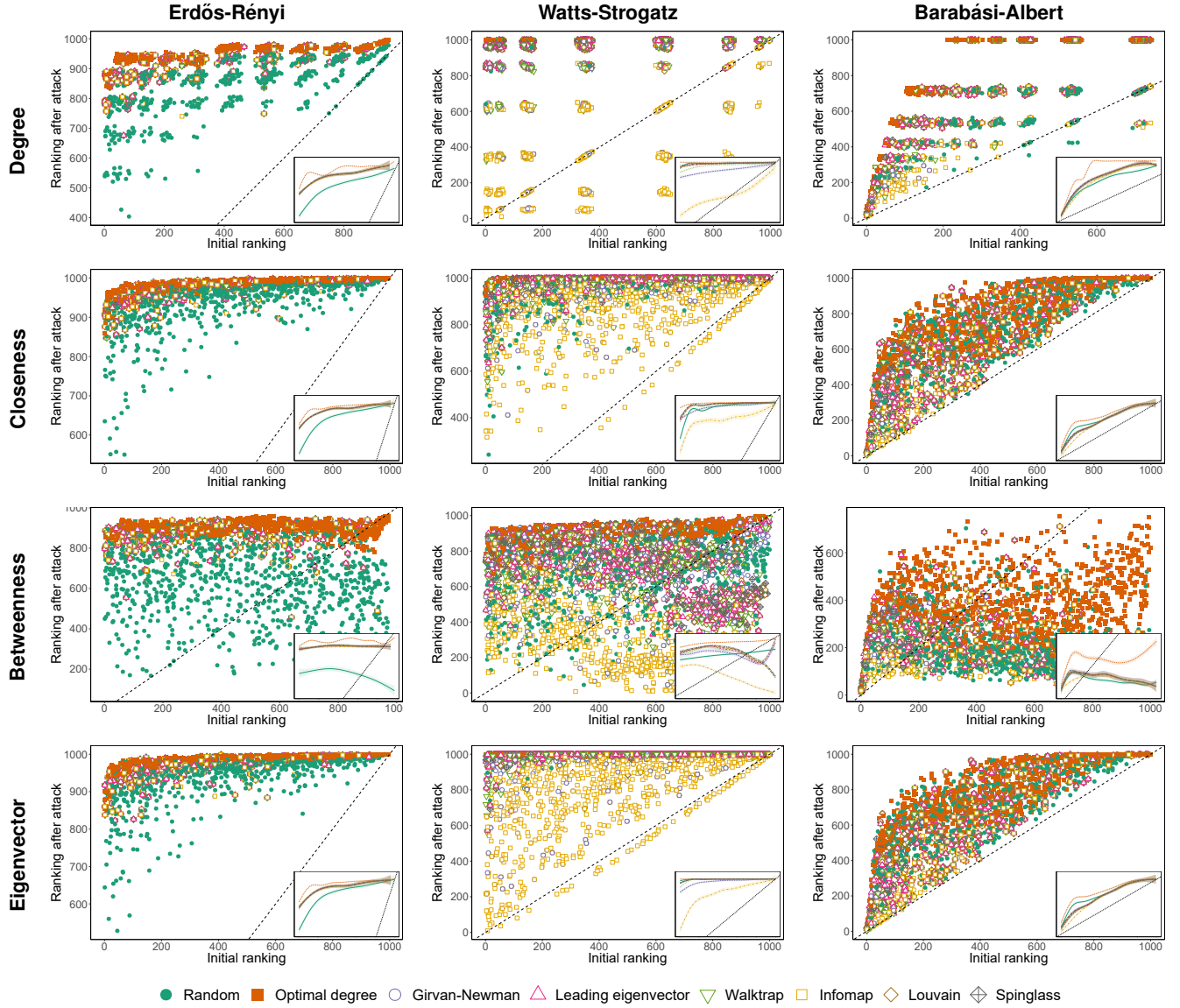


Figure 3: Effect of the evader’s initial ranking position on hiding effectiveness. Each row corresponds to a different centrality measure, while each column shows results for a different network generation model. In each scatter plot, the x -axis represents the evader’s initial ranking, and the y -axis represents the ranking of the most exposed Sybil. The dashed diagonal line marks equal rankings for both. Inset line plots show locally weighted polynomial regression fits with 95% confidence intervals.

5.3.3 Hiding in Covert Networks. We now turn to the experiments conducted on real-world covert networks. Here, our goal is to compare the Sybil attacks on centrality measures proposed in this work with previously studied hiding strategies based on edge modifications. In particular, we benchmark our Sybil-based methods against the ROAM (Remove One, Add Many) strategy [35], in which the evader disconnects from one of their alters and compensates by connecting that alter to one or more other alters.

In our experiments, we consider the simplest nontrivial Sybil attack scenario, where the evader, representing a leader of a covert

organization, splits their identity into two Sybils without creating a direct edge between them. We then compare the effectiveness of this setup with two executions of the $ROAM(k)$ heuristic, where k denotes the total number of modified edges in the network. In other words, $ROAM(k)$ removes one edge and adds $k - 1$ new ones.

Figure 4 summarizes the results of these experiments, where for each evader we report only the best-performing community-based Sybil strategy. As shown, in most cases the Sybil-based attacks outperform the ROAM heuristic by a substantial margin, despite maintaining the same total number of edges in the network, whereas

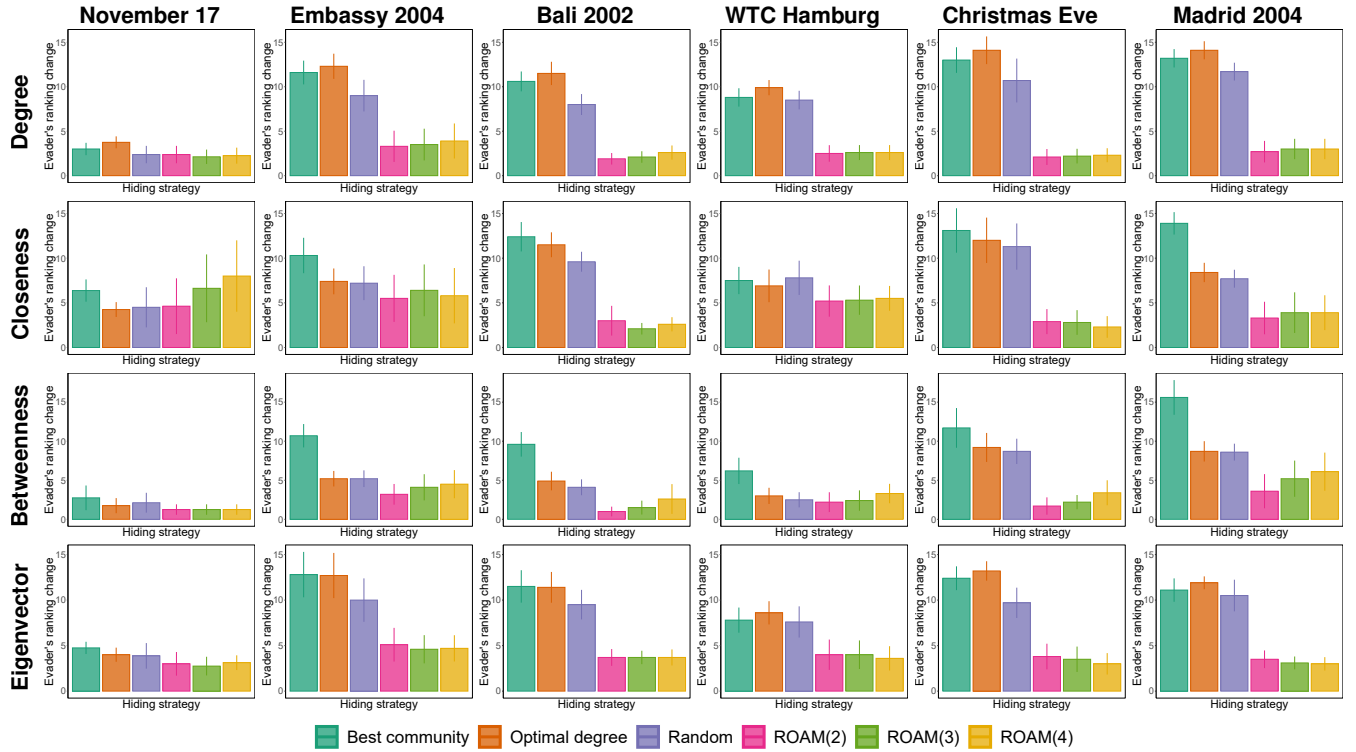


Figure 4: Results of hiding in real-life covert networks. Each row corresponds to a different centrality measure, and each column represents a different dataset. In each bar plot, the x -axis shows the hiding strategy, while the y -axis shows the change in ranking, i.e., either the difference between the initial ranking of the evader and the most exposed Sybil (for Sybil-based strategies), or between the initial and final rankings of the evader (for ROAM). Error bars indicate 95% confidence intervals.

high-budget ROAM variants actually increase network density. This outcome indicates that, within covert networks, Sybil attacks can pose a more serious and efficient threat to centrality-based analysis than traditional edge-rewiring approaches commonly explored in the literature.

6 Discussion

In this work, we conducted a comprehensive theoretical and empirical investigation of how Sybil attacks can be leveraged to manipulate centrality-based analyses. Formally, we demonstrated that identifying an optimal Sybil strategy is solvable in polynomial time for degree centrality, but is NP-complete for both closeness and betweenness centralities. This complexity gap underscores how even relatively simple network measures can become vulnerable under strategic manipulation. Experimentally, we showed that carefully designed Sybil attacks can substantially reduce an evader's apparent centrality across a wide spectrum of networks, ranging from synthetic models with diverse structural properties to real-world covert networks. In particular, both community-based heuristics and the optimal degree-centrality strategy consistently achieved large drops in ranking positions, often outperforming established edge-rewiring approaches from the literature.

Beyond their algorithmic significance, these findings carry broader implications for the analysis and governance of networked systems.

In domains such as counterterrorism, misinformation mitigation, and cybersecurity, analysts often rely on centrality measures to identify influential or critical actors. Our results reveal that such methods can be systematically deceived when individuals or entities can generate multiple identities, intentionally distorting structural signals. This highlights the need for increased methodological caution in interpreting centrality-based evidence and calls for analytical protocols that explicitly consider the possibility of Sybil manipulation. From a governance perspective, these findings further motivate the implementation of identity verification and cross-layer validation mechanisms to mitigate risks of structural deception in both social and technological networks.

Looking ahead, several promising directions for future research emerge. One avenue is the development of Sybil-resilient centrality measures, i.e., metrics that explicitly account for identity multiplicity or penalize implausible local structures. Another is the design of detection algorithms capable of identifying subgraphs or connectivity patterns characteristic of multi-identity evasion strategies. Extending this work to dynamic or multilayer networks, where Sybil identities evolve over time or act across interconnected systems, presents additional challenges. Finally, combining behavioral or contextual signals with structural analysis could enable hybrid approaches that preserve the interpretability of centrality metrics while enhancing their robustness against manipulation.

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A Supplementary Tables

Symbol	Meaning
V	The set of nodes
E	The set of edges
$N_G(v)$	The neighbors of v in G
$\delta_G(v)$	The degree of v in G
$d_G(v, w)$	The distance between v and w in G
$K_G(v)$	The nodes reachable from v in G
$\pi_G(v, w)$	The shortest paths between v and w in G
c_{deg}	The degree centrality measure
c_{clo}	The closeness centrality measure
c_{bet}	The betweenness centrality measure
c_{eig}	The eigenvector centrality measure
$\mathcal{A}$	A Sybil attack
$\mathcal{Q}$	The Sybils
$\mathcal{E}$	The edges incident with Sybils
$\emptyset$	An empty Sybil attack
$\mathcal{A} \cdot G$	The network resulting from the attack
$\mathcal{A} \cdot V$	The nodes resulting from the attack
$\mathcal{A} \cdot E$	The edges resulting from the attack
$v^\dagger$	The evader
b	The hiding budget
θ	The safety threshold

Table 2: Summary of the notation.

B The Proof of Theorem 2

PROOF. The problem is clearly in NP since after performing a Sybil attack we can compute the ranking of the closeness centrality in polynomial time and validate the solution.

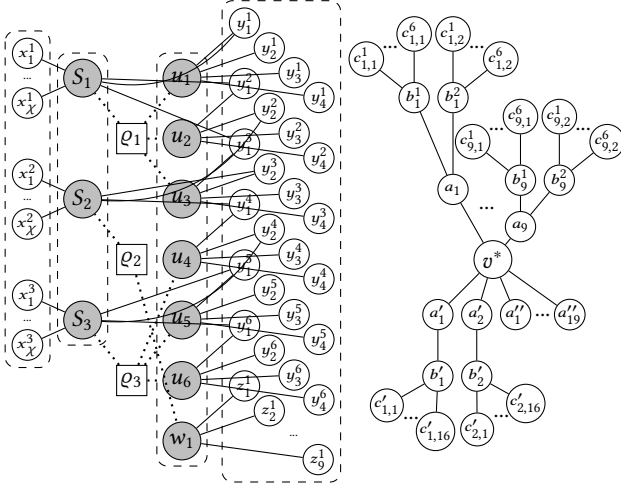


Figure 5: Example of the construction used in the proof of Theorem 2. In the example $k = 2$. The alters are grey, the Sybils are the square nodes, and the edges added to the Sybils are dotted. Each group of nodes encircled with a dashed frame is additionally connected into a clique.

Dataset	$ V $	$ E $	Potential evaders
November 17	18	46	8
Embassy	27	112	23
Bali 2002	27	158	25
WTC Hamburg	32	121	18
Christmas Eve	45	234	38
Madrid 2004	54	226	39

Table 3: Structural properties of the covert network datasets.

We will prove that the problem is NP-hard using a reduction from an NP-complete Exact 3-Set Cover problem [19]. The problem involves a set $S = \{S_1, \dots, S_{|S|}\}$ of subsets of a universe U , and a constant $k \in \mathbb{N}$ such that $\forall S_i \in S (S_i \subseteq U \wedge |S_i| = 3)$ and $|U| = 3k$. The goal is to identify a k -subset of S that covers the entire U , i.e., $S^* \subseteq S$ such that $\bigcup_{S_i \in S^*} S_i = U$ and $|S^*| = k$.

Let $\mathcal{I}_C = (S, U, k)$ be an arbitrary instance of the Exact 3-Set Cover problem. We will slightly abuse the notation by using S to denote $|S|$ in the numerical formulas. Assume that $S \geq 3$ and $k \geq 3$, all smaller instances can be solved via exhaustive search in polynomial time. In what follows, let $\zeta_j^i = \{j' \leq j : u_i \in S_{j'}\}$, and let $\chi = 4S^2 + 5k$. We construct the network $G = (V, E)$ in the following way (an example of this construction is presented in Figure 5, with the network G after the Sybil attack occurred):

- $V = S \cup U \cup W \cup X \cup Y \cup Z \cup A \cup B \cup C \cup \{v^*, v^*\}$,
- $E = (\{v^*\} \times (S \cup U \cup W)) \cup E_1 \cup E_2 \cup E_3$,

where

- $W = \{w_i\}_{i=1}^{S-k}$,
- $X = \bigcup_{i=1}^S \{x_j^i\}_{j=1}^{4S^2+5k}$,
- $Y = \bigcup_{i=1}^{3k} \{y_j^i\}_{j=1}^{S+1}$,
- $Z = \bigcup_{i=1}^{S-k} \{z_j^i\}_{j=1}^{3S}$,

- $A = \{a_i\}_{i=1}^{S^2} \cup \{a'_i\}_{i=1}^k \cup \{a''_i\}_{i=1}^{3S+k+8}$,
- $B = \{b_i^1, b_i^2\}_{a_i \in A} \cup \{b'_i\}_{i=1}^k \cup \{b''_i\}_{i=1}^{2k-9}$,
- $C = \bigcup_{b'_i \in B} \{c_{i,j}^h\}_{h=1}^{2S} \cup \bigcup_{b'_i \in B} \{c'_{i,j}\}_{j=1}^{5S}$,
- $E_1 = \{(S_j, y_{\zeta_j^i}^i)\}_{u_i \in S_j} \cup \{(x_j^i, S_i)\}_{x_j^i \in V}$
 $\cup \{(y_j^i, u_i)\}_{y_j^i \in V: i \leq 3k} \cup \{(y_j^i, w_{i-3k})\}_{y_j^i \in V: i > 3k}$,

i.e., the nodes $S \cup U$ form a structure corresponding to the given instance of the Exact 3-Set Cover problem, with nodes $W \cup X \cup Y$ serving as intermediaries,

- $E_2 = \{(S_i, S_j)\}_{S_i, S_j \in S} \cup \{(u_i, u_j)\}_{u_i, u_j \in U}$
 $\cup \{(u_i, w_j)\}_{u_i \in U, w_j \in W} \cup \{(w_i, w_j)\}_{w_i, w_j \in W}$
 $\cup \{(x_j^i, x_{j'}^{i'})\}_{x_j^i, x_{j'}^{i'} \in X} \cup \{(y_j^i, y_{j'}^{i'})\}_{y_j^i, y_{j'}^{i'} \in Y}$
 $\cup \{(y_j^i, z_{j'}^{i'})\}_{y_j^i \in Y, z_{j'}^{i'} \in Z} \cup \{(z_j^i, z_{j'}^{i'})\}_{z_j^i, z_{j'}^{i'} \in Z}$,

i.e., the sets of nodes $S, U \cup W, X$, and $Y \cup Z$ are each connected into a clique,

- $E_3 = \{(a_i, v^*)\}_{a_i \in A} \cup \{(a'_i, v^*)\}_{a'_i \in A} \cup \{(a''_i, v^*)\}_{a''_i \in A}$
 $\cup \{(b_i^j, a_i)\}_{b_i^j \in B} \cup \{(b'_i, a'_i)\}_{b'_i \in B} \cup \{(c_{i,j}^h, b_i^j)\}_{c_{i,j}^h \in C}$
 $\cup \{(c'_{i,j}, b'_i)\}_{c'_{i,j} \in C}$,

i.e., the nodes $A \cup B \cup C$ form a tree with v^* as the root.

We will call the component of G containing S and U the main component, and the one containing the node v^* the tree component.

We now construct an instance $\mathcal{I}_S = (G, v^*, c, b, \theta)$ of the Central-ity Sybil Attack problem, where:

- $G = (V, E)$ is the network constructed above,
- v^* is the evader,
- $c = c_{clo}$ is the closeness centrality,
- $b = S$ is the hiding budget,
- $\theta = 4S^3 + 3S^2 + 5Sk + S + 3k + 1$ is the safety threshold.

In what follows, let $D_{\mathcal{A}}(v, H) = \sum_{h \in (K_{\mathcal{A}, G}(v) \cap H) \setminus v} d_{\mathcal{A}, G}(v, h)$, and let $D^* = 12S^3 + 5S^2 + 15Sk + 3S + 8k - 9$.

LEMMA 1. *Let $\mathcal{A} = (Q, \mathcal{E})$ be a Sybil attack of size b fulfilling points 1 and 2 of Definition 2. There are exactly $4S^3 + 3S^2 + 5Sk + S + 3k + 1$ non-Sybil nodes in $\mathcal{A} \cdot G$ such that $D_{\mathcal{A}}(v, V \cup Q) < D^*$, and there exists a non-Sybil node such that $D_{\mathcal{A}}(v, V \cup Q) = D^* - 1$. Hence, $\mathcal{A}$ is a solution to $\mathcal{I}_S$ if and only if $\forall \varrho \in Q D_{\mathcal{A}}(\varrho, V \cup Q) \geq D^*$.*

PROOF. First, notice that after the attack both components of $\mathcal{A} \cdot G$ have the same number of nodes, i.e., $4S^3 + 3S^2 + 3S + 5Sk + 5k$. Therefore, the closeness centrality of any $v \in V \cup Q$ depends solely on $D_{\mathcal{A}}(v, V \cup Q)$, and for a given $v \in V$ and any $\varrho \in Q$ we have

$$c(\mathcal{A} \cdot G, v) > c(\mathcal{A} \cdot G, \varrho) \Leftrightarrow D_{\mathcal{A}}(v, V \cup Q) < D_{\mathcal{A}}(\varrho, V \cup Q).$$

Table 4 presents distances between nodes in the main component of $\mathcal{A} \cdot G$. From the values in the table we get:

- $D_{\mathcal{A}}(S_i, V \cup Q) \leq 8S^3 + 3S^2 + 10kS + 4S + k - 4 < D^*$,
- $D_{\mathcal{A}}(x_i, V \cup Q) \leq 4S^3 + 10S^2 + 5kS + 6S + 9k - 5 < D^*$,
- $D_{\mathcal{A}}(u_i, V \cup Q) \geq 12S^3 + 6S^2 + 15kS + 3S + 8k - 2 > D^*$,
- $D_{\mathcal{A}}(w_i, V \cup Q) \geq 12S^3 + 6S^2 + 15kS + S + 8k - 1 > D^*$,
- $D_{\mathcal{A}}(y_i, V \cup Q) \leq 12S^3 + 4S^2 + 15kS + 4S + 7k - 2 < D^*$,
- $D_{\mathcal{A}}(z_i, V \cup Q) \leq 12S^3 + 4S^2 + 15kS + 4S + 7k - 2 < D^*$.

For the tree component we have (remember $S \geq 3$ and $k \geq 3$):

- $D_{\mathcal{A}}(v^*, V \cup Q) = 12S^3 + 5S^2 + 15Sk + 3S + 8k - 10 = D^* - 1 < D^*$,
- $D_{\mathcal{A}}(a_i, V \cup Q) \geq 16S^3 + 8S^2 + 20Sk - 2S + 13k - 16 > D^*$,
- $D_{\mathcal{A}}(a'_i, V \cup Q) \geq 16S^3 + 8S^2 + 10Sk + 6S + 4k + 15 > D^*$,
- $D_{\mathcal{A}}(a''_i, V \cup Q) = 16S^3 + 8S^2 + 20Sk + 6S + 13k - 12 > D^*$,

v	$D_{\mathcal{A}}(v, S)$	$D_{\mathcal{A}}(v, X)$	$D_{\mathcal{A}}(v, U)$	$D_{\mathcal{A}}(v, W)$	$D_{\mathcal{A}}(v, Y)$	$D_{\mathcal{A}}(v, Z)$	$D_{\mathcal{A}}(v, Q)$
S_i	$S - 1$	$(2S - 1)\chi$	$\leq 3k$	$\leq 3(S - k)$	$6k(S + 1) - 3$	$6S(S - k)$	$\leq S^2$
x_i	$2S - 1$	$S\chi - 1$	$\leq 4k$	$\leq 4(S - k)$	$9k(S + 1) - 3$	$9S(S - k)$	$\leq S^2$
u_i	$\geq 2S$	$\geq 3S\chi$	$3k - 1$	$S - k$	$(6k - 1)(S + 1)$	$6S(S - k)$	$\geq S$
w_i	$\geq 2S$	$\geq 3S\chi$	$3k$	$S - k - 1$	$6k(S + 1)$	$3S(2(S - k) - 1)$	$\geq S$
y_i	$\leq 2S$	$\leq 3S\chi$	$6k - 1$	$2(S - k)$	$3k(S + 1) - 1$	$3S(S - k)$	$\leq S^2$
z_i	$2S$	$\leq 3S\chi$	$6k$	$2(S - k) - 1$	$3k(S + 1)$	$3S(S - k) - 1$	$\leq S^2$

Table 4: The distances between the nodes in the main component of $\mathcal{A} \cdot G$.

- $D_{\mathcal{A}}(b_i^j, V \cup Q) \geq 12S^3 + 7S^2 + 25Sk + 5S + 18k - 14 > D^*$,
- $D_{\mathcal{A}}(b_i', V \cup Q) \geq 20S^3 + 11S^2 + 25Sk - 11S + 3k + 26 > D^*$,
- $D_{\mathcal{A}}(b_i'', V \cup Q) \geq 20S^3 + 11S^2 + 25Sk + 7k + 2 > D^*$,
- $D_{\mathcal{A}}(c_{i,j}^h, V \cup Q) \geq 16S^3 + 4S^2 + 30Sk + 4S + 23k - 10 > D^*$,
- $D_{\mathcal{A}}(c_{i,j}^l, V \cup Q) \geq 24S^3 + 14S^2 + 30Sk - 8S + 4k + 33 > D^*$.

As can be seen, there are $7S^2 + S + 8k + 1$ non-Sybil nodes in $\mathcal{A} \cdot G$ such that $D_{\mathcal{A}}(v, V \cup Q) < D^*$, and these are the nodes in S (S nodes), X ($4S^3 + 5Sk$ nodes), Y ($3Sk + 3k$ nodes), Z ($3S^2 - 3Sk$ nodes), and the node v^* . Moreover, we have that $D_{\mathcal{A}}(v^*, V \cup Q) = D^* - 1$. This concludes the proof of the first claim of the lemma.

As for the second claim, notice that for $\mathcal{I}_S$ we have $\theta = 7S^2 + S + 8k + 1$. Hence, if $\forall_{\varrho \in Q} D_{\mathcal{A}}(\varrho, V \cup Q) \geq D^*$ then all nodes in $\{v^*\} \cup S \cup X \cup Y \cup Z$ contribute to θ , and $\mathcal{A}$ is indeed a solution to $\mathcal{I}_S$. To prove the other implication, let $\mathcal{A}$ be an arbitrary solution to $\mathcal{I}_S$. Since θ is satisfied, there are at least $\theta = 7S^2 + S + 8k + 1$ nodes $v \in V$ such that $D_{\mathcal{A}}(v, V \cup Q)$ is strictly smaller than for any of the Sybil nodes. Since v^* is one of the $7S^2 + S + 8k + 1$ nodes with the smallest $D_{\mathcal{A}}(v, V \cup Q)$, we must have that $\forall_{\varrho \in Q} D_{\mathcal{A}}(\varrho, V \cup Q) > D_{\mathcal{A}}(v^*, V \cup Q) = D^* - 1$. Therefore, $\forall_{\varrho \in Q} D_{\mathcal{A}}(\varrho, V \cup Q) \geq D^*$. Hence, $\mathcal{A}$ is a solution to $\mathcal{I}_S$ if and only if $\forall_{\varrho \in Q} D_{\mathcal{A}}(\varrho, V \cup Q) \geq D^*$. This concludes the proof of the lemma. $\square$

Consider $\varrho \in Q$. Let S_{ϱ} denote the number of S_i that ϱ is connected to in $\mathcal{A} \cdot G$. Similarly, let U_{ϱ} and W_{ϱ} denote the number u_i and w_i that ϱ is connected to in $\mathcal{A} \cdot G$. Finally, let $\Delta_{\varrho} = |\{(\varrho, u_i) \in \mathcal{E} : \neg \exists_{S_j} (u_i \in S_j \wedge (\varrho, S_j) \in \mathcal{E})\}|$, i.e., the number of nodes u_i connected to ϱ such that ϱ is not connected to any S_j containing u_i .

LEMMA 2. Let $\mathcal{A} = (Q, \mathcal{E})$ be a Sybil attack of size b fulfilling points 1 and 2 of Definition 2. $\mathcal{A}$ is a solution to $\mathcal{I}_S$ if and only if $\forall_{\varrho \in Q} (S_{\varrho} = 1 \wedge ((U_{\varrho} = 3 \wedge \Delta_{\varrho} = 0 \wedge W_{\varrho} = 0) \vee (U_{\varrho} = 0 \wedge W_{\varrho} = 1)))$.

PROOF. Table 5 presents distances from the Sybils to the other nodes in the main component of $\mathcal{A} \cdot G$, which will prove to be useful for computing the value of $D_{\mathcal{A}}(\varrho, V \cup Q)$.

We begin with proving the first implication of the lemma. Assume that $\mathcal{A}$ is a solution to $\mathcal{I}_S$. Consider a Sybil node $\varrho \in Q$:

- Assume $S_{\varrho} \geq 1 \wedge U_{\varrho} + W_{\varrho} = 0$. If $S_{\varrho} > 1$ then:

$$\begin{aligned} D_{\mathcal{A}}(\varrho, V \cup Q) &\leq 12S^3 + 3S^2 + 15Sk + 6S + 7k - 8 \\ &< 12S^3 + 5S^2 + 15Sk + 3S + 8k - 9 = D^*. \end{aligned}$$

From Lemma 1 such $\mathcal{A}$ cannot be a solution to $\mathcal{I}_S$. Hence, if $\mathcal{A}$ is a solution and $S_{\varrho} \geq 1 \wedge U_{\varrho} + W_{\varrho} = 0$, then $S_{\varrho} \leq 1$.

- Assume $S_{\varrho} \geq 1 \wedge U_{\varrho} + W_{\varrho} \geq 1$. If $S_{\varrho} > 1$ then:

$$\begin{aligned} D_{\mathcal{A}}(\varrho, V \cup Q) &\leq 12S^3 + 3S^2 + 15Sk + 3S + 3k - 4 \\ &< 12S^3 + 5S^2 + 15Sk + 3S + 8k - 9 = D^*. \end{aligned}$$

From Lemma 1 such $\mathcal{A}$ cannot be a solution to $\mathcal{I}_S$. Therefore, if $\mathcal{A}$ is a solution and $S_{\varrho} \geq 1 \wedge U_{\varrho} + W_{\varrho} \geq 1$, then $S_{\varrho} \leq 1$.

We have that if $\mathcal{A}$ is a solution to $\mathcal{I}_S$, and $S_{\varrho} \geq 1$ then $S_{\varrho} \leq 1$. Hence, each Sybil can be connected to at most one node from S . However, there are as many Sybils as nodes in S , and since $\mathcal{A}$ is a solution, each node in S has a Sybil neighbor. Therefore, each Sybil is connected to exactly one node in S .

Notice that $D_{\mathcal{A}}(\varrho, V \cup Q)$ is minimized if we do not add any edges between Sybils. In that case we get the equation we mark $(\star)$:

$$D_{\mathcal{A}}(\varrho, V \cup Q) = 12S^3 + 5S^2 + 15Sk + 6S + 8k - 3 - \Delta_{\varrho} - U_{\varrho}(S + 2) - W_{\varrho}(3S + 1).$$

Assume $W_{\varrho} > 1$. From $(\star)$ we have $D_{\mathcal{A}}(\varrho, V \cup Q) \leq 12S^3 + 5S^2 + 15Sk + 8k - 8 < D^*$. From Lemma 1 such $\mathcal{A}$ cannot be a solution.

Therefore if ϱ is connected to any node from W , then $W_{\varrho} = 1$. In such case from $(\star)$ we get $D_{\mathcal{A}}(\varrho, V \cup Q) = 12S^3 + 5S^2 + 15Sk + 3S + 8k - 4 - \Delta_{\varrho} - U_{\varrho}(S + 2)$.

If $U_{\varrho} > 0$ then $D_{\mathcal{A}}(\varrho, V \cup Q) \leq 12S^3 + 5S^2 + 15Sk + 2S + 8k - 6 < D^*$. Again, from Lemma 1 such $\mathcal{A}$ cannot be a solution. Therefore if $W_{\varrho} = 1$ then $U_{\varrho} = 0$, which gives us $D_{\mathcal{A}}(\varrho, V \cup Q) = 12S^3 + 5S^2 + 15Sk + 3S + 8k - 7 > D^*$, thus fulfilling the condition from Lemma 1.

Hence, $S - k$ of the Sybils are connected to one node from W each, and to no nodes from U . Therefore, the remaining k Sybils have to be connected with all nodes from U .

Assume that $U_{\varrho} > 3$. From $(\star)$ we then have $D_{\mathcal{A}}(\varrho, V \cup Q) \leq 12S^3 + 5S^2 + 15Sk + 2S + 8k - 11 < D^*$. From Lemma 1 such $\mathcal{A}$ cannot be a solution, hence $U_{\varrho} \leq 3$. However, since k Sybils have to be connected with all $3k$ nodes from U , we must have $U_{\varrho} = 3$ for such nodes. From $(\star)$ we then have $D_{\mathcal{A}}(\varrho, V \cup Q) = 12S^3 + 5S^2 + 15Sk + 3S + 8k - 9 - \Delta_{\varrho} = D^* - \Delta_{\varrho}$. Hence, from Lemma 1 we have that such $\mathcal{A}$ is a solution only if $\Delta_{\varrho} = 0$. This concludes the proof of the first implication.

To prove the other implication, let $\mathcal{A} = (Q, \mathcal{E})$ be a Sybil attack of size b fulfilling points 1 and 2 of Definition 2, and let

$$\forall_{\varrho \in Q} (S_{\varrho} = 1 \wedge ((U_{\varrho} = 3 \wedge \Delta_{\varrho} = 0 \wedge W_{\varrho} = 0) \vee (U_{\varrho} = 0 \wedge W_{\varrho} = 1))).$$

Take Sybil node ϱ such that $S_{\varrho} = 1 \wedge U_{\varrho} = 3 \wedge \Delta_{\varrho} = 0 \wedge W_{\varrho} = 0$. From the crucial equation we have $D_{\mathcal{A}}(\varrho, V \cup Q) = 12S^3 + 5S^2 + 15Sk + 3S + 8k - 9 = D^*$.

On the other hand, take Sybil node ϱ such that $U_{\varrho} = 0 \wedge W_{\varrho} = 1$ (notice that for such a node $\Delta_{\varrho} = 3$). We then have $D_{\mathcal{A}}(\varrho, V \cup Q) = 12S^3 + 5S^2 + 15Sk + 3S + 8k - 7 > D^*$.

Therefore, all Sybil nodes fulfill the requirements of Lemma 1, and $\mathcal{A}$ is a solution to $\mathcal{I}_S$. This concludes the proof of the lemma. $\square$

We now show that $\mathcal{I}_S$ has a solution if and only if $\mathcal{I}_C$ has one.

Assumption	$D_{\mathcal{A}}(\varrho, S)$	$D_{\mathcal{A}}(\varrho, X)$	$D_{\mathcal{A}}(\varrho, U)$	$D_{\mathcal{A}}(\varrho, W)$	$D_{\mathcal{A}}(\varrho, Y)$	$D_{\mathcal{A}}(\varrho, Z)$	$D_{\mathcal{A}}(\varrho, Q)$
$U_{\varrho} + W_{\varrho} = 0$	$2S - S_{\varrho}$	$(3S - S_{\varrho})\chi$	$12k - 3$	$4S - 4k$	$9Sk + 9k - 3$	$9S^2 - 9Sk$	$\leq 2S^2$
$U_{\varrho} + W_{\varrho} \geq 1$	$2S - S_{\varrho}$	$(3S - S_{\varrho})\chi$	$6k - U_{\varrho}$	$2S - 2k - W_{\varrho}$	$9Sk + 9k - \Delta_{\varrho} - U_{\varrho}(S+1)$	$9S^2 - 9Sk - 3W_{\varrho}S$	$\leq 2S^2$

Table 5: The distances from Sybils to the other nodes in the main component of $\mathcal{A} \cdot G$, assuming that $S_{\varrho} \geq 1$.

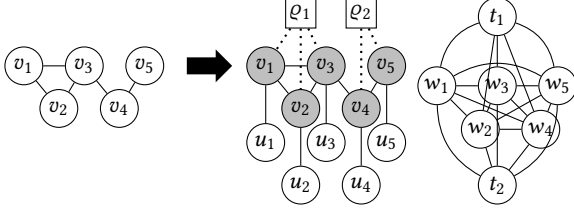


Figure 6: Example of the construction used in the proof of Theorem 3. The alters are grey, the Sybils are the square nodes, and the edges added to the Sybils are dotted.

To prove the first implication assume that $\mathcal{I}_S$ has a solution $\mathcal{A} = (Q, \mathcal{E})$. Let $S^* = \bigcup_{u_i \in U} \{S_j \in S : \exists \varrho \in Q(\varrho, u_i) \in \mathcal{E} \wedge (\varrho, S_j) \in \mathcal{E}\}$. From point 2 of Definition 2 we have that each $u_j \in U$ have a neighbor in Q . Moreover, from Lemma 2 we have that if $(\varrho, u_i) \in \mathcal{E}$ then there is $(\varrho, S_j) \in \mathcal{E}$ (since $S_{\varrho} = 1$) and $u_i \in S_j$ (since $\Delta_{\varrho} = 0$). Hence, S^* is a cover of U . Moreover, since there are $S - k$ nodes W , and each $\varrho \in Q$ is connected to at most one node in W (as either $W_{\varrho} = 0$ or $W_{\varrho} = 1$), there are at least $S - k$ nodes $\varrho \in Q$ with $U_{\varrho} = 0$. Therefore, there are at most k nodes in $\varrho \in Q$ with $U_{\varrho} > 0$, and these are exactly the nodes in S^* . Therefore S^* is a k -cover of U and a solution to $\mathcal{I}_C$.

To prove the other implication assume that $\mathcal{I}_C$ has a solution $S^* \subseteq S$. Let $\mathcal{A} = (Q, \mathcal{E})$ be a Sybil attack where:

- $Q = \{\varrho_1, \dots, \varrho_S\}$,
- $\mathcal{E} = \bigcup_{S_i \in S^*} \left(\{(\varrho_i, S_i)\} \cup \{(\varrho_i, u_j)\}_{u_j \in S_i} \right) \cup \bigcup_{S_i \notin S^*} \left(\{(\varrho_i, S_i), (\varrho_i, w_{|j \leq i: S_j \notin S^*|})\} \right)$.

$\mathcal{A}$ is of size b and fulfills points 1 and 2 of Definition 2, as:

- (1) $\forall \varrho_i \in Q d_{\mathcal{A} \cdot V}(\varrho_i, S_i) = 1 < \infty$,
- (2) $\forall v \in S \cup U \cup W \exists \varrho_i \in Q(\varrho_i, v) \in \mathcal{E}$, since every $S_i \in S$ is connected with ϱ_i , every $u_i \in U$ is connected with ϱ_j such that $S_j \in S^*$ and $u_i \in S_j$ (there must exist such S_j since S^* is a cover of U), and for each $S_i \notin S^*$ (there are $S - k$ such nodes) ϱ_i is connected with one of the $S - k$ nodes from W .

Therefore, from Lemma 2 we have that $\mathcal{A}$ is the solution to $\mathcal{I}_S$. $\square$

C The Proof of Theorem 3

PROOF. The problem is clearly in NP since after performing a Sybil attack we can compute the ranking of the betweenness centrality in polynomial time and validate the solution.

We will now prove that the problem is NP-hard. To this end we will use a reduction from the Clique Cover problem, one of the Karp's original NP-complete problems [21]. The problem involves a network $G_C = (V_C, E_C)$ and a constant $k_C \in \mathbb{N}$, with the goal of identifying at most k_C cliques induced in G_C that cover V_C .

Let $\mathcal{I}_C = (G_C, k_C)$ be an arbitrary instance of the Clique Cover problem, and let $n_C = |V_C|$. We construct the network G in the

following way (an example of this construction is presented in Figure 6, with the network G after the Sybil attack occurred):

- $V = V_C \cup \{v^\dagger, t_1, t_2\} \cup \bigcup_{v_i \in V_C} \{u_i\} \cup \bigcup_{i=1}^{n_C+k_C-2} \{w_i\}$,
- $E = E_C \cup (\{v^\dagger\} \times V_C) \cup \bigcup_{v_i \in V_C} \{(v_i, u_i)\} \cup \bigcup_{w_i \in V} \{(w_i, t_1), (w_i, t_2)\} \cup \bigcup_{w_i, w_j \in V} \{(w_i, w_j)\}$.

We now construct an instance $\mathcal{I}_S = (G, v^\dagger, c, b, \theta)$ of the Central-ity Sybil Attack problem, where:

- $G = (V, E)$ is the network constructed above,
- $v^\dagger$ is the evader,
- $c = c_{bet}$ is the betweenness centrality,
- $b = k_C$ is the hiding budget,
- $\theta = 2n_C + k_C - 2$ is the safety threshold.

Consider $\mathcal{A} = (Q, \mathcal{E})$, an arbitrary Sybil attack on network G . The betweenness centrality values for the non-Sybil nodes in the network $\mathcal{A} \cdot G$ resulting from the attack are as follows:

- $c_{bet}(\mathcal{A} \cdot G, u_i) = c_{bet}(\mathcal{A} \cdot G, t_1) = c_{bet}(\mathcal{A} \cdot G, t_2) = 0$, as these nodes control no shortest paths,
- $c_{bet}(\mathcal{A} \cdot G, w_i) = \frac{1}{n_C+k_C-2}$, as each node w_i controls one of the shortest paths between t_1 and t_2 , with other paths running through other nodes w_j ,
- $c_{bet}(\mathcal{A} \cdot G, v_i) \geq 2n_C - 2$, as each node v_i controls all shortest paths between u_i and other nodes u_j and v_j .

Given the above, only v_i and w_i can contribute towards the safety threshold, as all other nodes have the minimal possible betweenness centrality value of zero. Moreover, since $\theta = 2n_C + k_C - 2$, and since there are exactly $2n_C + k_C - 2$ nodes v_i and w_i , all of them need to have greater betweenness centrality than all of the Sybils.

Consider a Sybil ϱ_i and the nodes from V_C that it is connected to as a result of the attack $\mathcal{A}$. If at least two of these nodes v_j, v_l are not connected with an edge from E_C we have that: $c_{bet}(\mathcal{A} \cdot G, \varrho_i) \geq \frac{1}{n_C+k_C-2}$, as ϱ_i controls a shortest path between v_i and v_j , and the other shortest paths between them can run through the other nodes in V_C and Q . Therefore the nodes w_i contribute towards θ if only if for every Sybil ϱ_i its neighbors in $\mathcal{A} \cdot G$ induce a clique.

We now show that $\mathcal{I}_S$ has a solution if and only if $\mathcal{I}_C$ has one.

Assume that $\mathcal{I}_S$ has a solution $\mathcal{A} = (Q, \mathcal{E})$. Since every w_i has to contribute to the safety threshold, we have that the neighbors of every $\varrho_i \in Q$ induce a clique in $\mathcal{A} \cdot G$. Let $V_i = V_C \cap N_{\mathcal{A} \cdot G}(\varrho_i)$. Since V_i induces a clique in G , and the edges between V_C in $\mathcal{A} \cdot G$ are the same as in G_C , then V_i also induces a clique in G_C . Moreover, since $\mathcal{A}$ is a solution to $\mathcal{I}_S$ and $N_G(v^\dagger) = V_C$, we have that $\bigcup_{\varrho_i \in Q} V_i = V_C$. Therefore $V_1, \dots, V_{k_C}$ is a solution to $\mathcal{I}_C$.

Now assume that $\mathcal{I}_C$ has a solution, i.e., $V_1, \dots, V_{k_C}$ such that $\bigcup_i V_i = V_C$ and each V_i induces a clique in G_C . Let $\mathcal{A} = (Q, \mathcal{E})$ be a Sybil attack where $Q = \{\varrho_1, \dots, \varrho_{k_C}\}$ and $\mathcal{E} = \bigcup_{\varrho_i \in Q} (\{\varrho_i\} \times V_i)$. Based on the above, for every Sybil node ϱ_i its neighbors in $\mathcal{A} \cdot G$ induce a clique, resulting in $c_{bet}(\mathcal{A} \cdot G, \varrho_i) = 0$. Therefore, all nodes w_i and v_i contribute towards the safety threshold, and $\mathcal{A}$ is the solution of $\mathcal{I}_S$. This concludes the proof. $\square$